

1	2	3	4	5	
A					A
B					B
C					C
D					D
1	2	3	4	5	

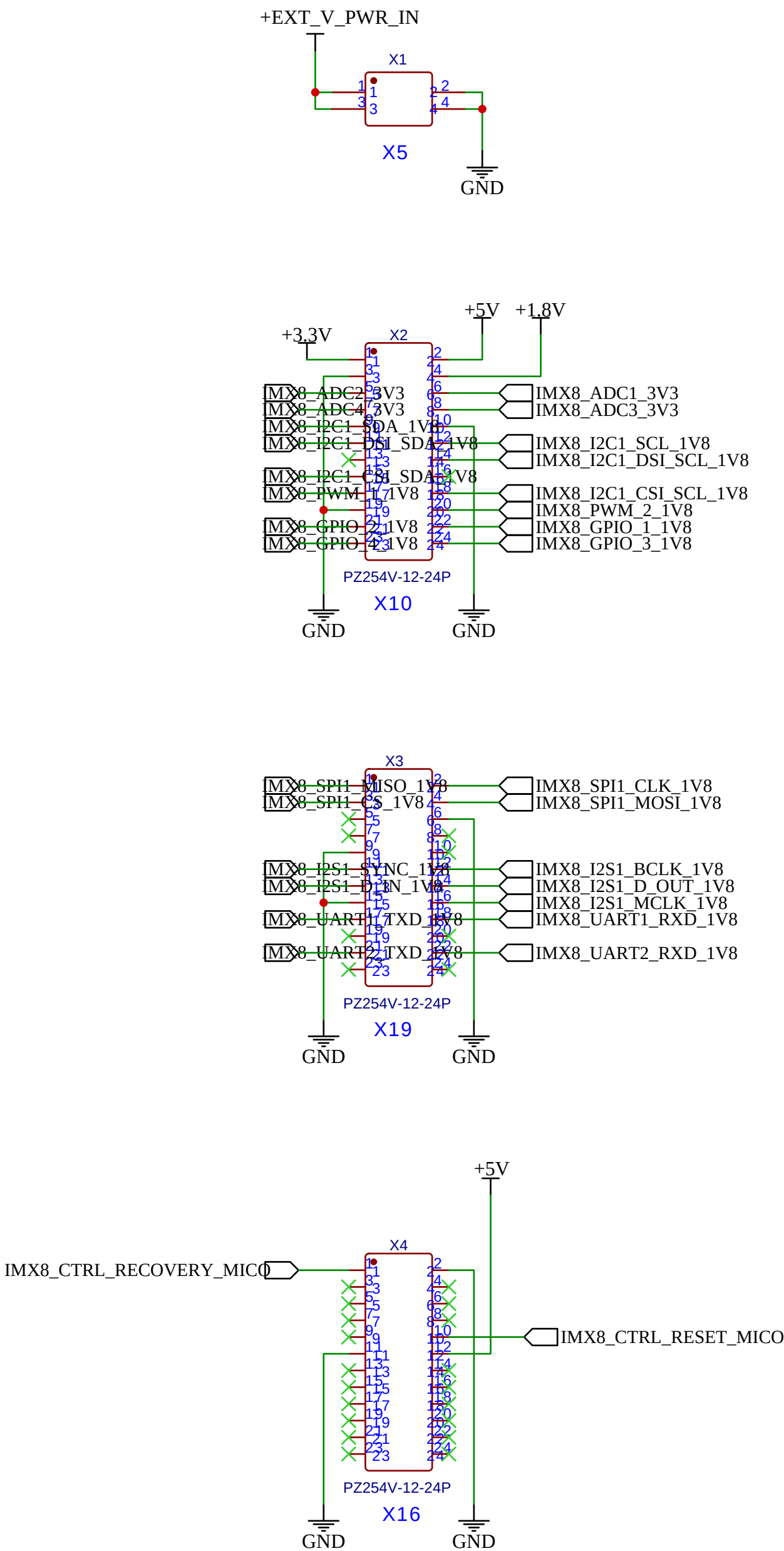
Toradex Mallow Carrier Board Expansion

TITLE: Mallow Carrier Board Expansion			REV: 1.2
	Company: CoreKinect		Sheet:1/15
	Date: 12/05/25 Drawn By: Mateo Segura		

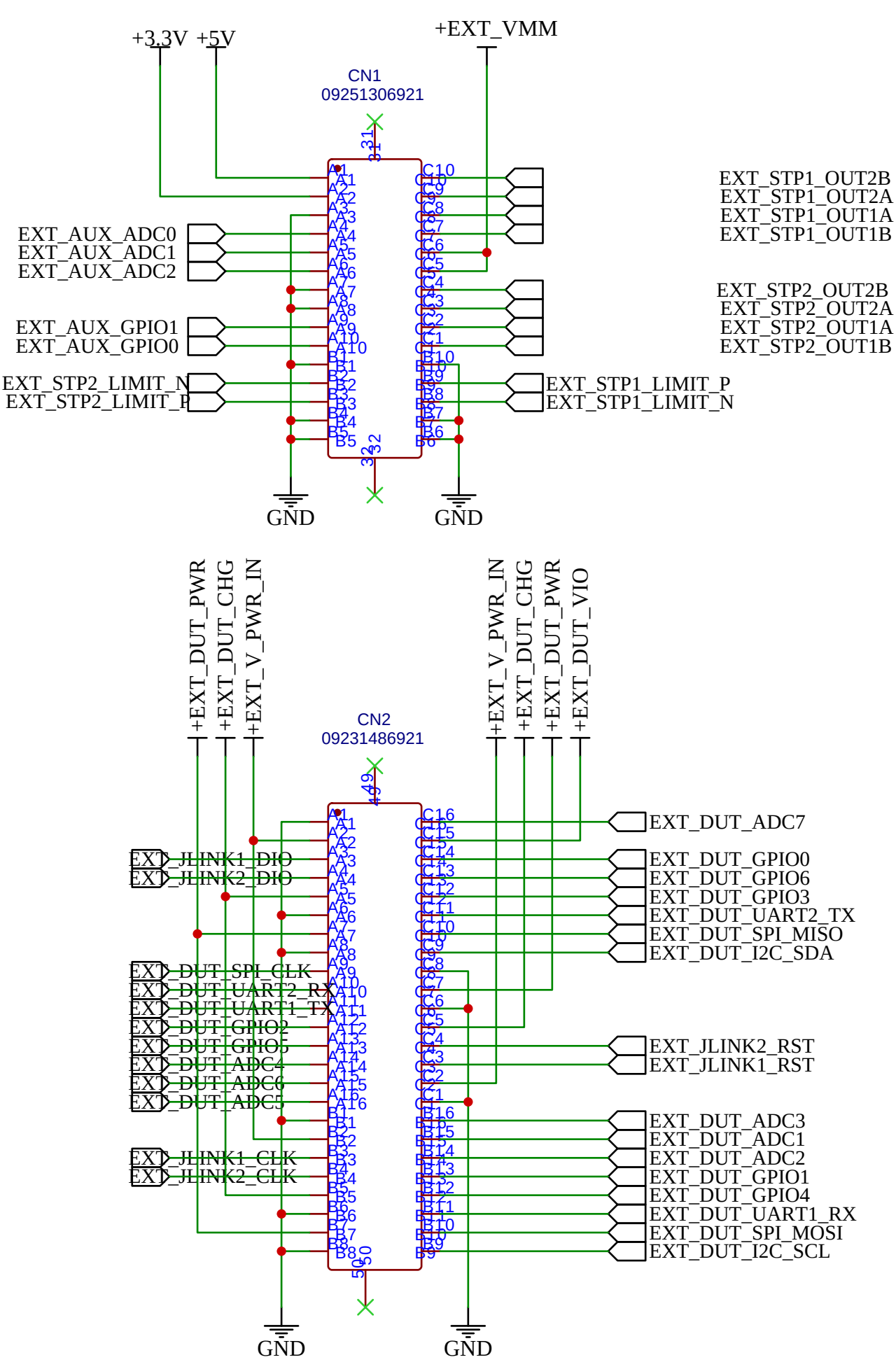
1	2	3	4	5
A				A
B				B
C				C
D				D
1	2	3	4	5

TITLE:	System Overview	REV: 1.2
	Company: CoreKinect	Sheet:2/15
Date: 12/05/25	Drawn By: Mateo Segura	

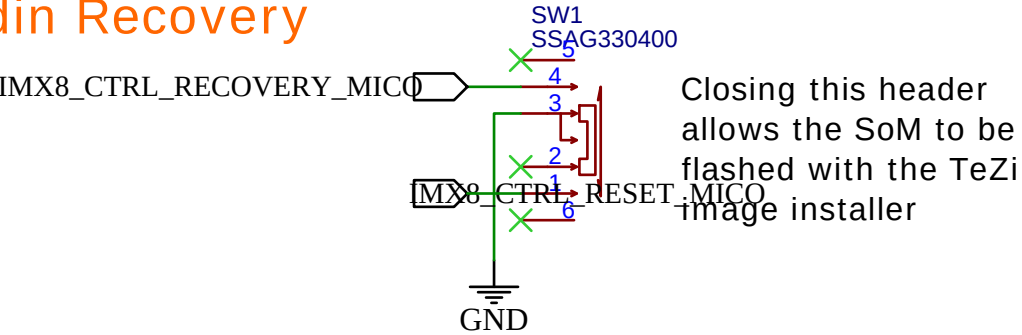
Mallow Carrier Board Connections



External Connectors



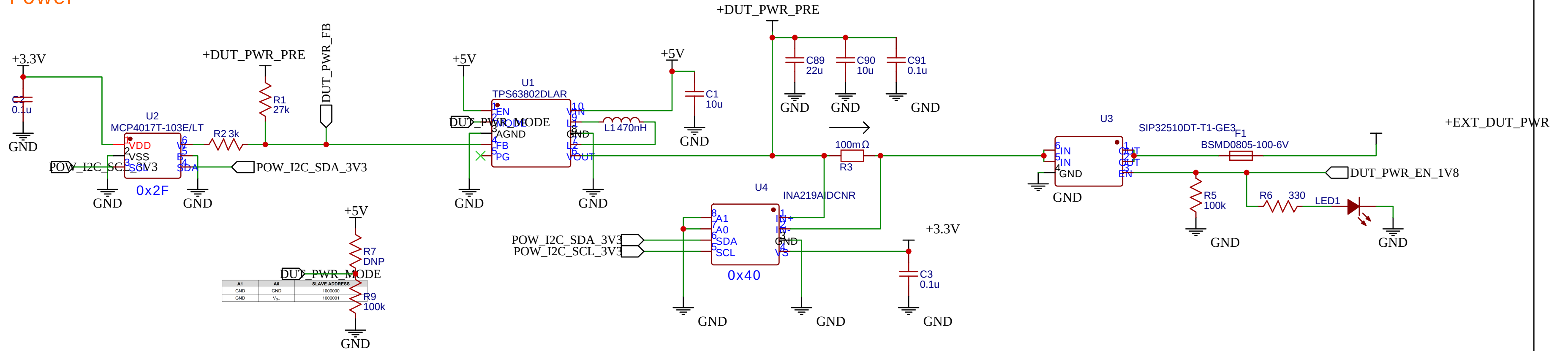
Verdin Recovery



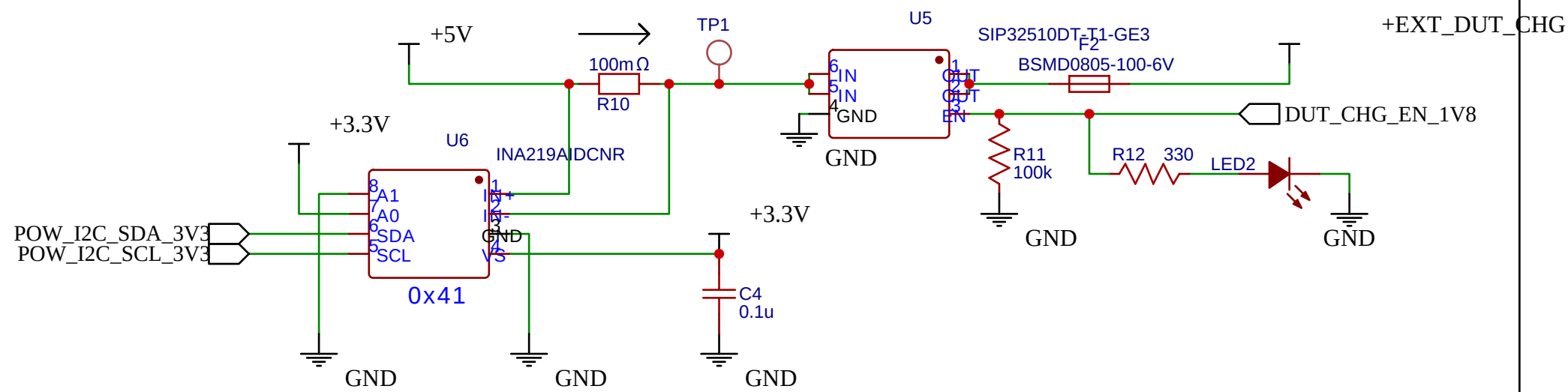
Closing this header allows the SoM to be flashed with the TeZi image installer

TITLE:		Connectors	REV: 1.2
COREKINET	Company: CoreKinect		Sheet:3/15
	Date: 12/05/25	Drawn By: Mateo Segura	

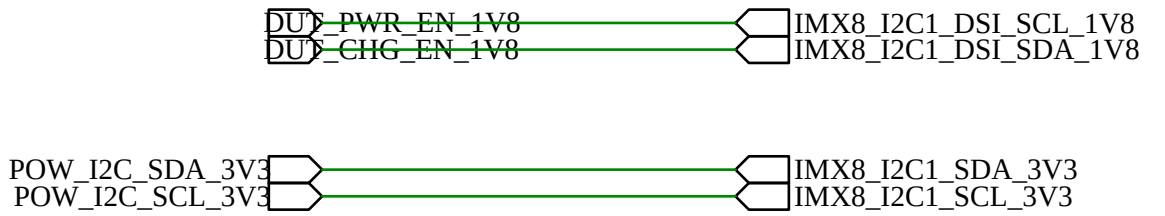
DUT Power



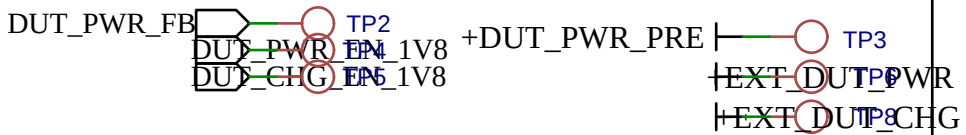
DUT Charge Power

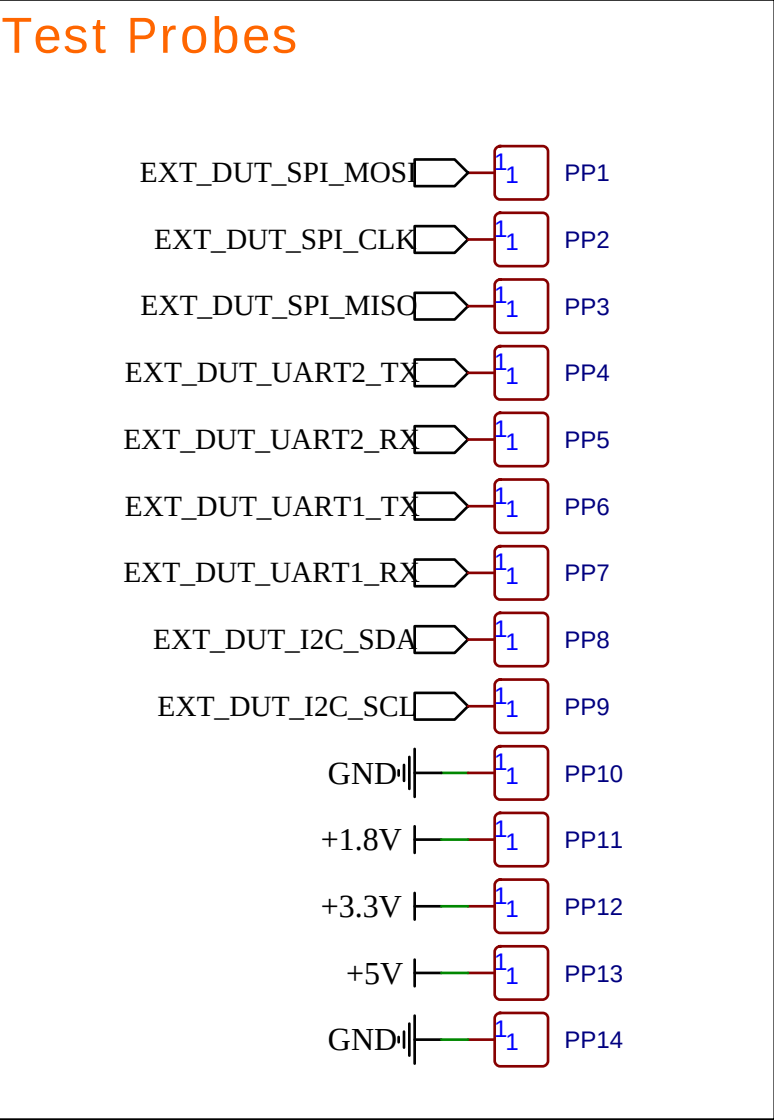
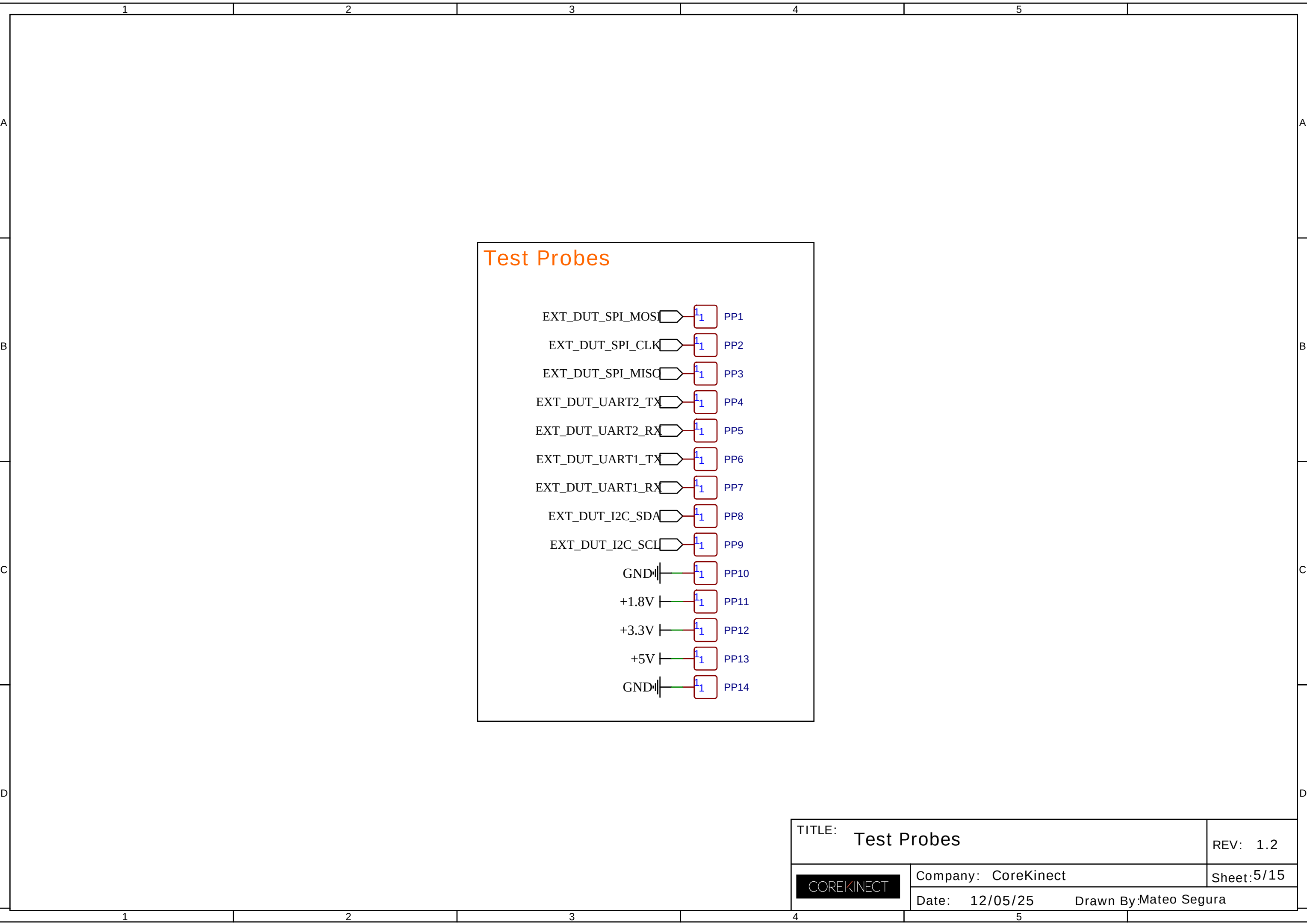


Internal Connections



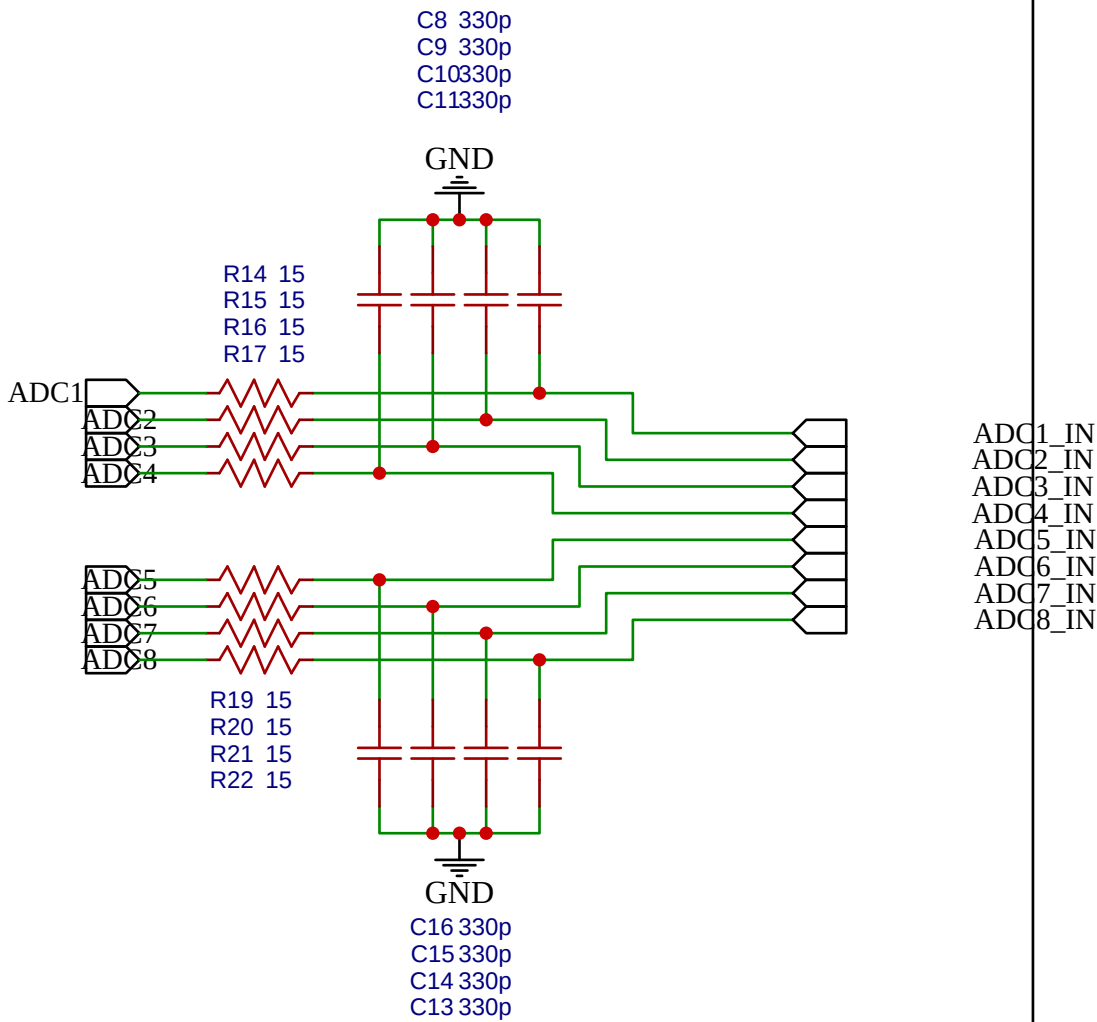
Test Points





TITLE: Test Probes		REV: 1.2
COREKINET	Company: CoreKinect	Sheet:5/15
	Date: 12/05/25 Drawn By: Mateo Segura	

Final Stage Signal Conditioning



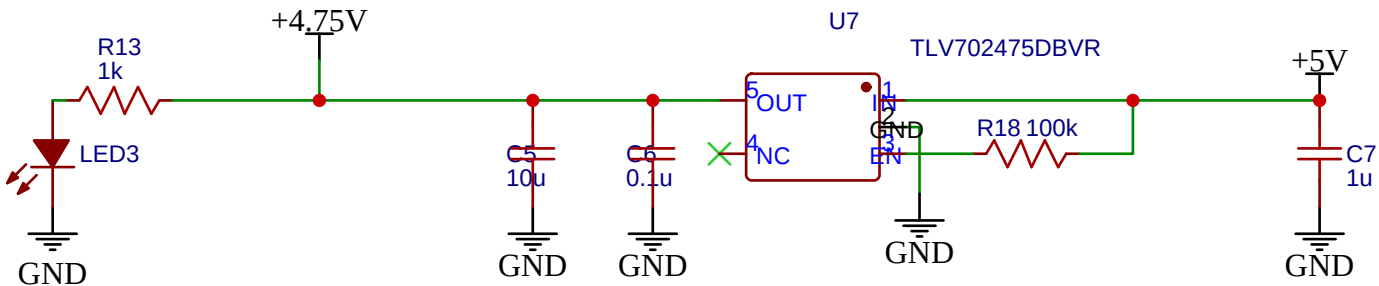
Internal Connections

ADC_I2C_SDA_3V3 BMX8_I2C1_SDA_3V3
ADC_I2C_SCL_3V3 BMX8_I2C1_SCL_3V3

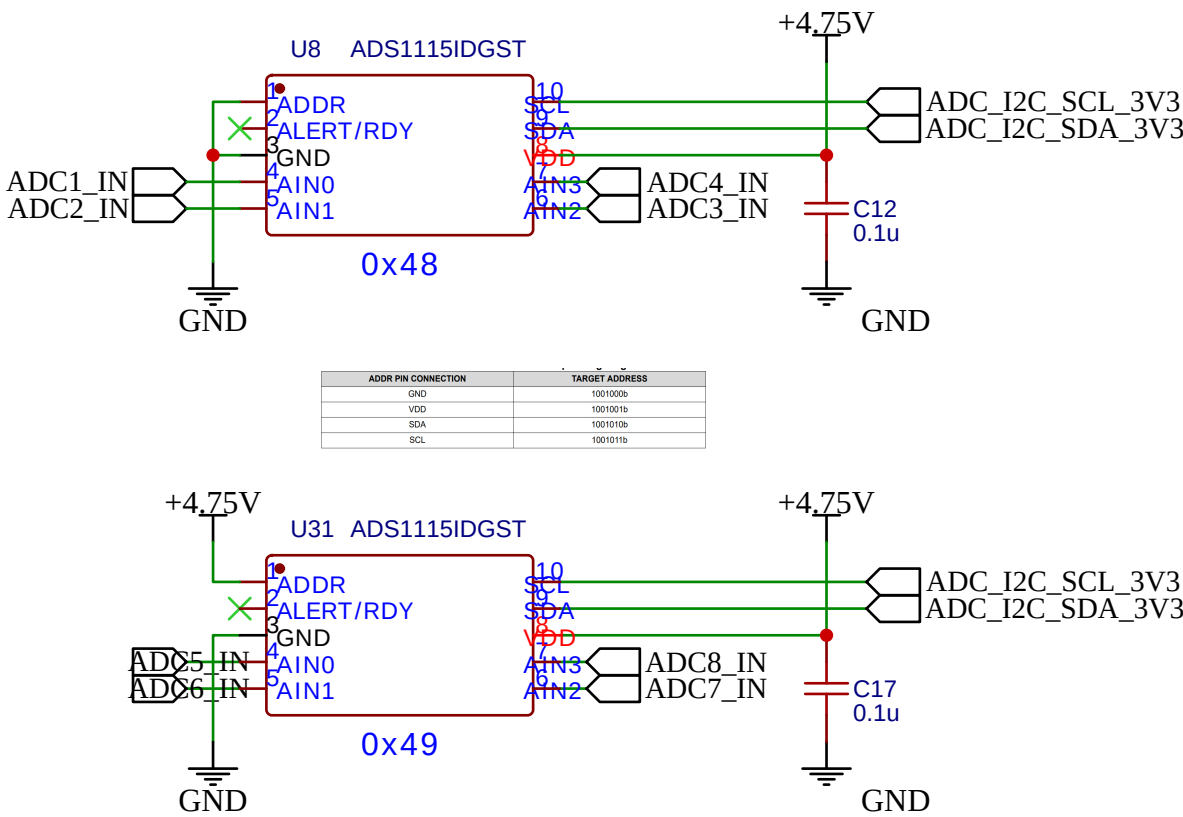
Test Points

TP12 +4.75V

Clean Power Supply



Analog to Digital Converters



TITLE: Analog to Digital Converter

REV: 1.2

COREKINET

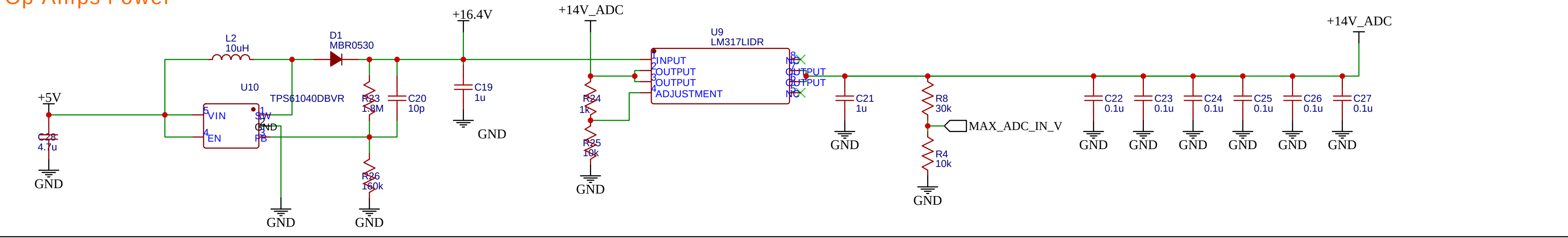
Company: CoreKinect

Sheet:6/15

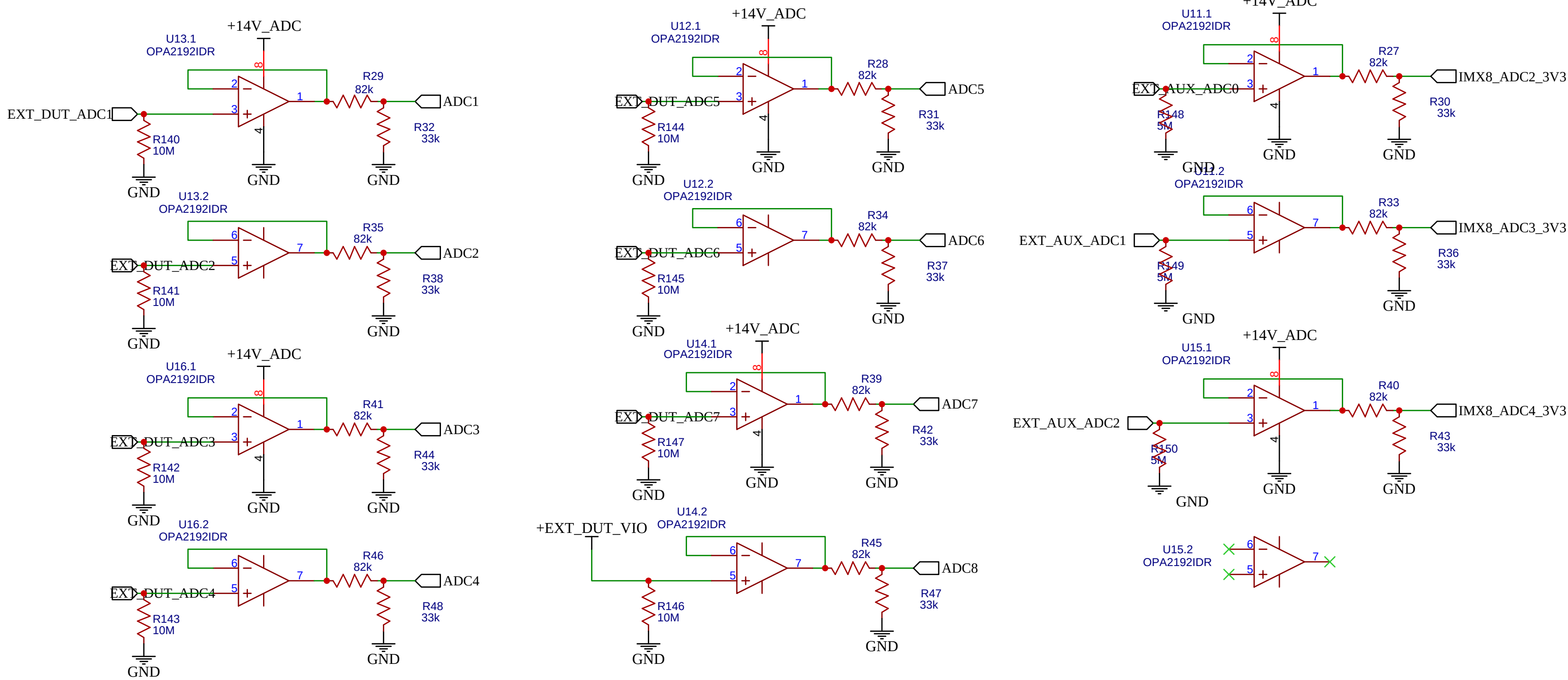
Date: 12/05/25

Drawn By: Mateo Segura

Op Amps Power



Signal Conditioning



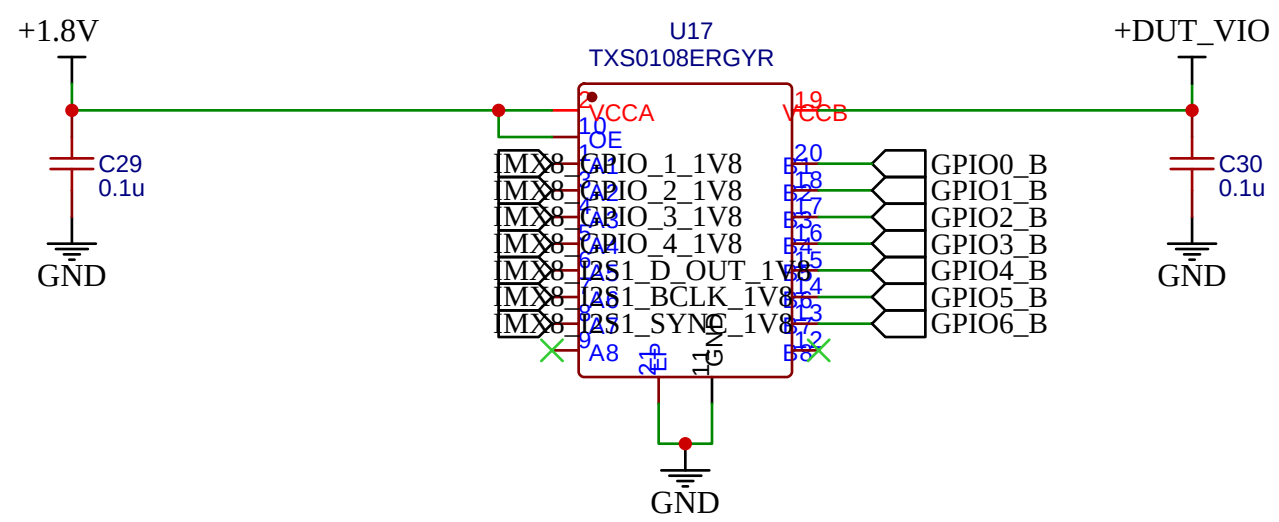
Internal Connections

MAX_ADC_IN_V IMX8_ADC1_3V3

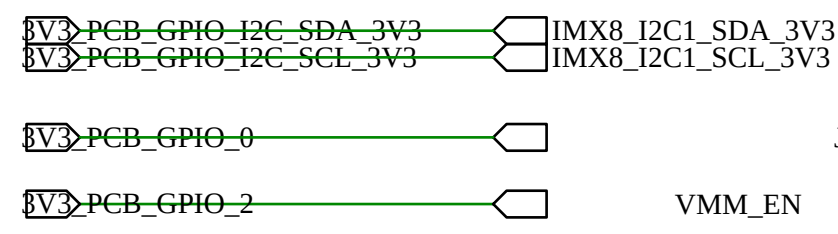
Test Points

TP16 16.4V
TP17 14V_ADC

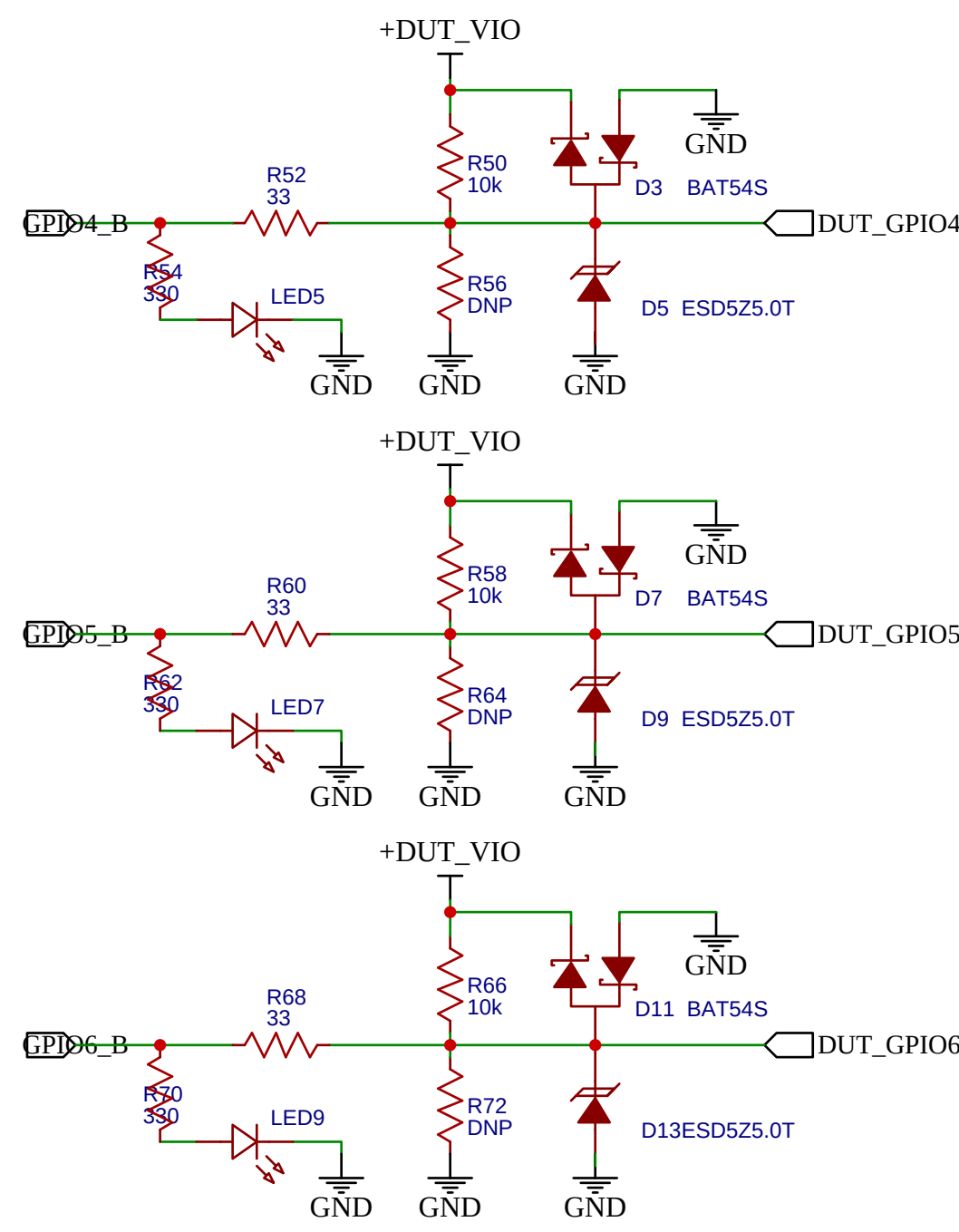
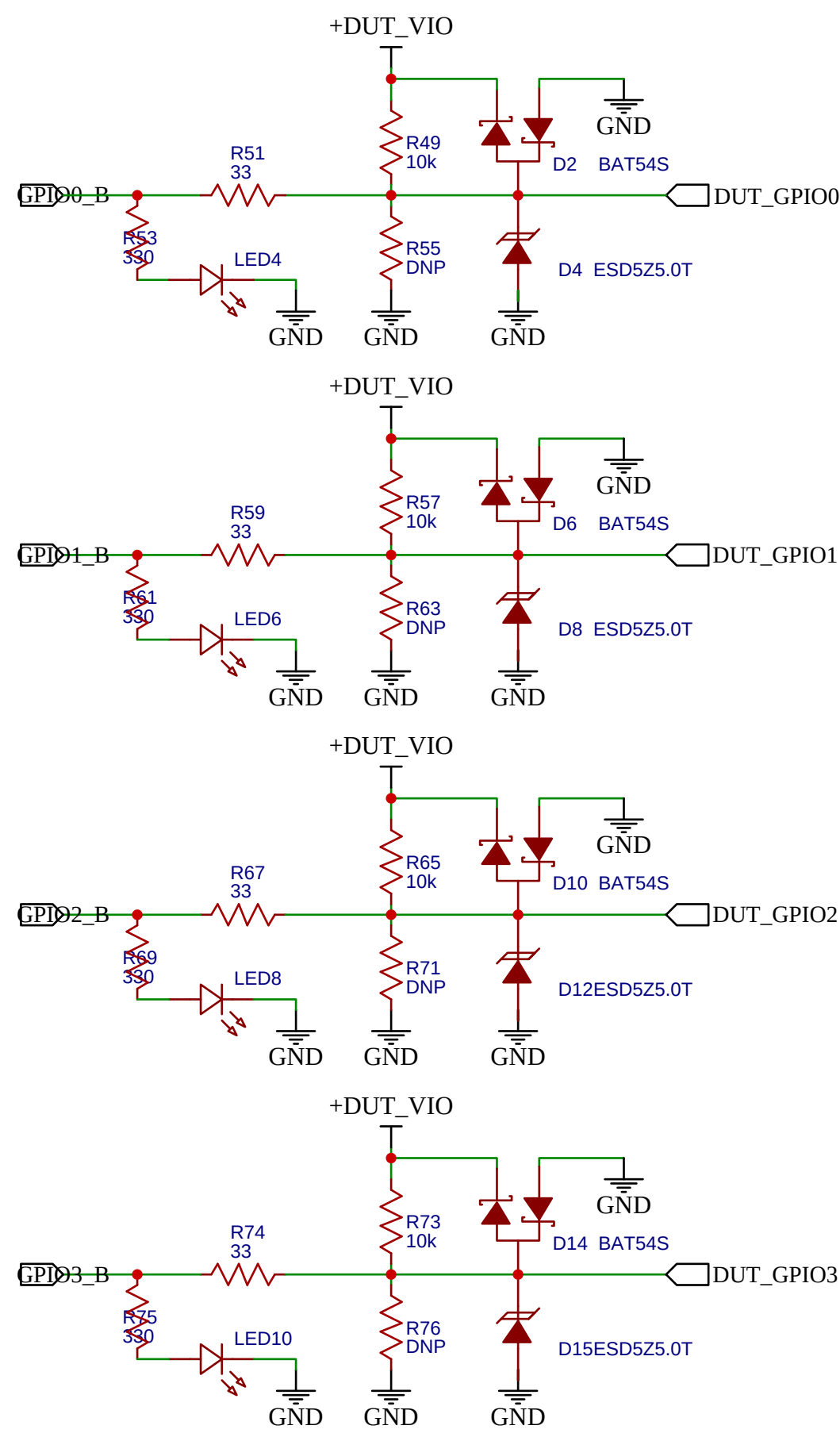
Level Shifter



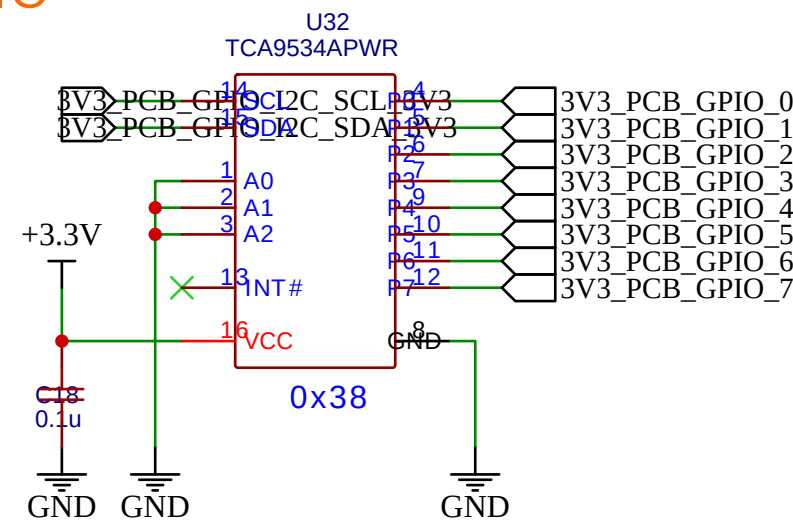
Internal Connections



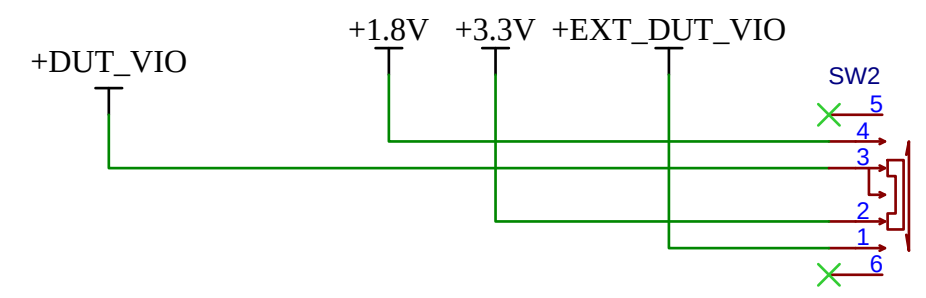
Signal Conditioning



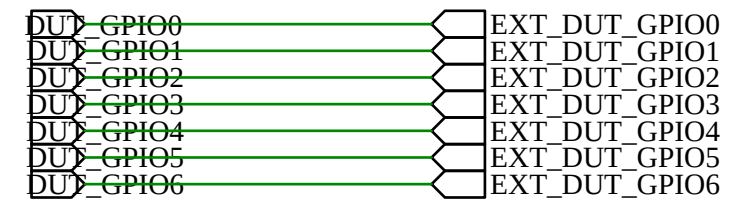
Extra PCB GPIO



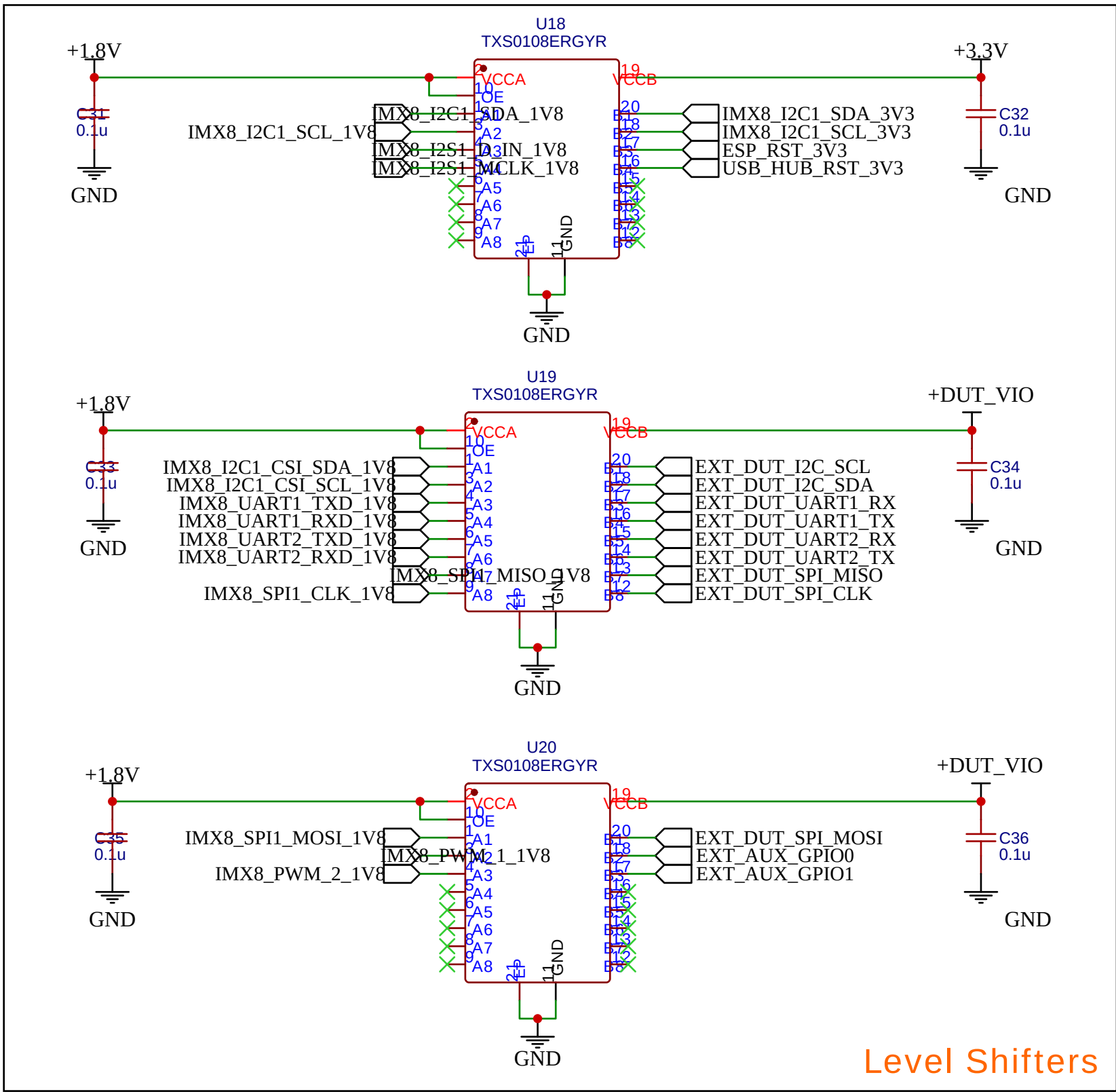
VIO Selector



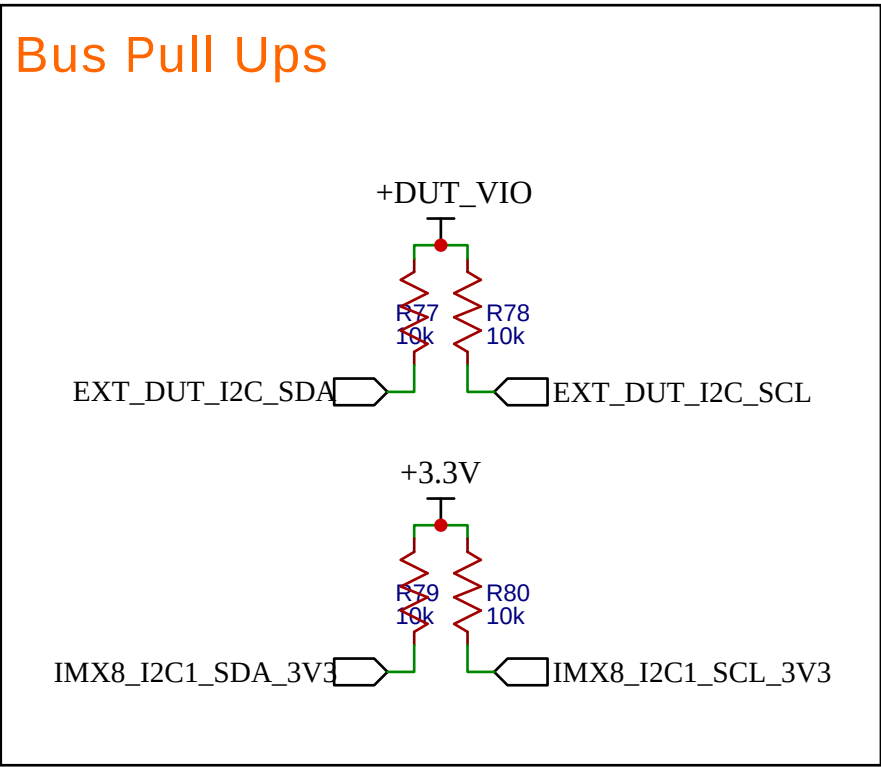
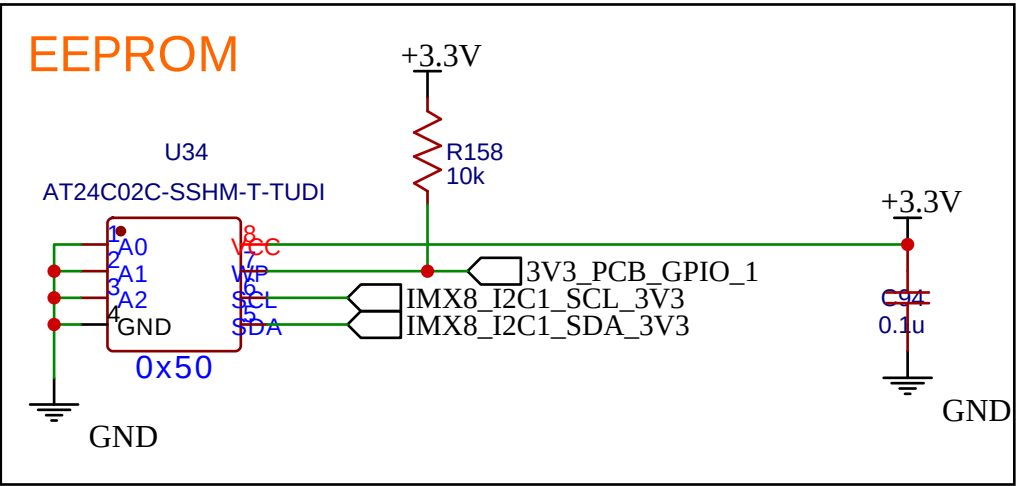
External Connections



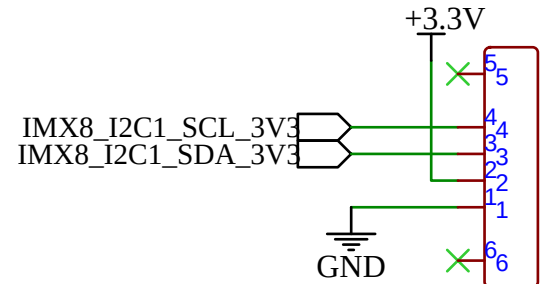
TITLE: DUT GPIO		REV: 1.2
COREKINECT	Company: CoreKinect	Sheet:8/15
	Date: 12/05/25	Drawn By: Mateo Segura



Level Shifters

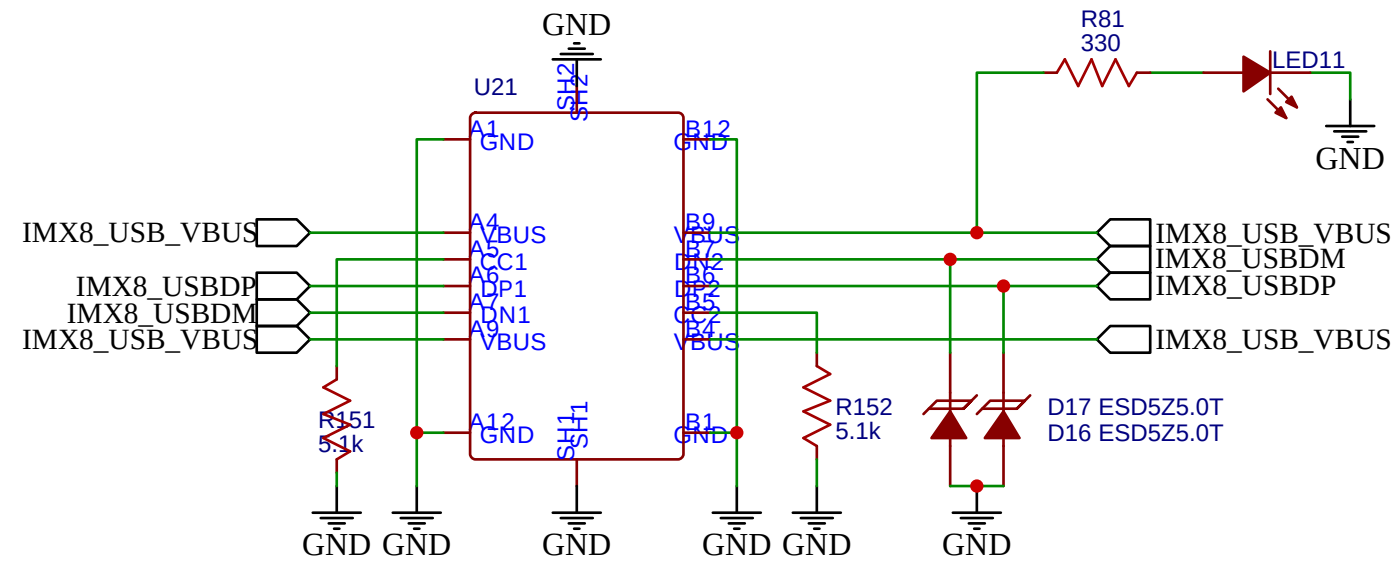


QWICC I2C Connector

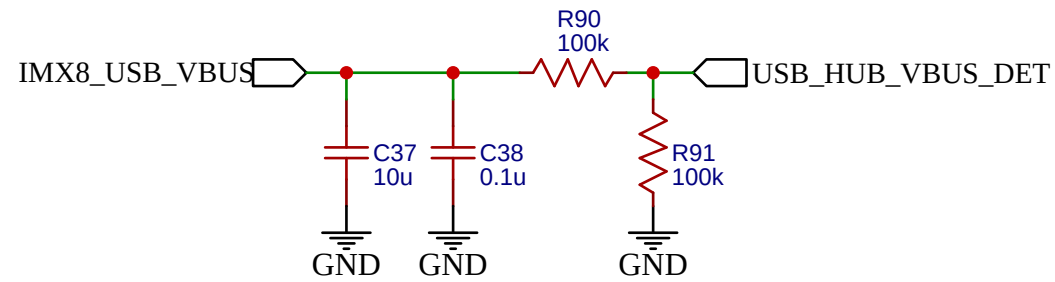


TITLE: Analog to Digital Converter		REV: 1.2
COREKINET	Company: CoreKinect	Sheet:9/15
	Date: 12/05/25	Drawn By: Mateo Segura

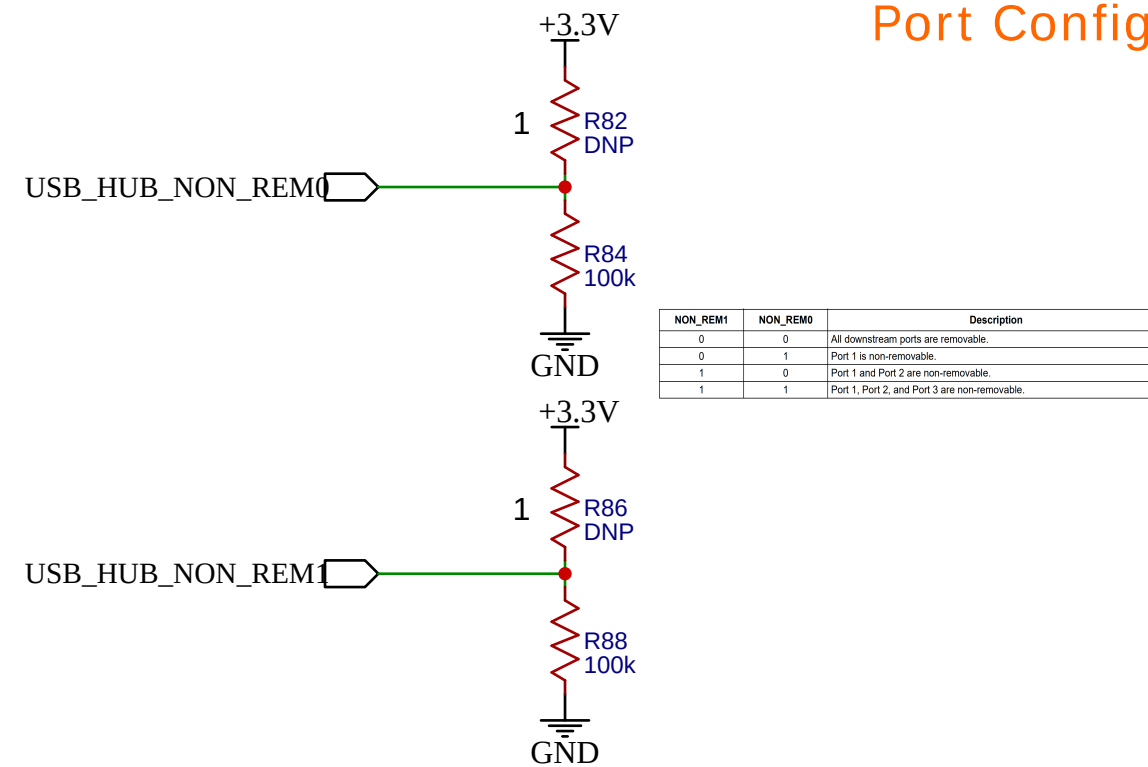
iMX8 Upstream Port



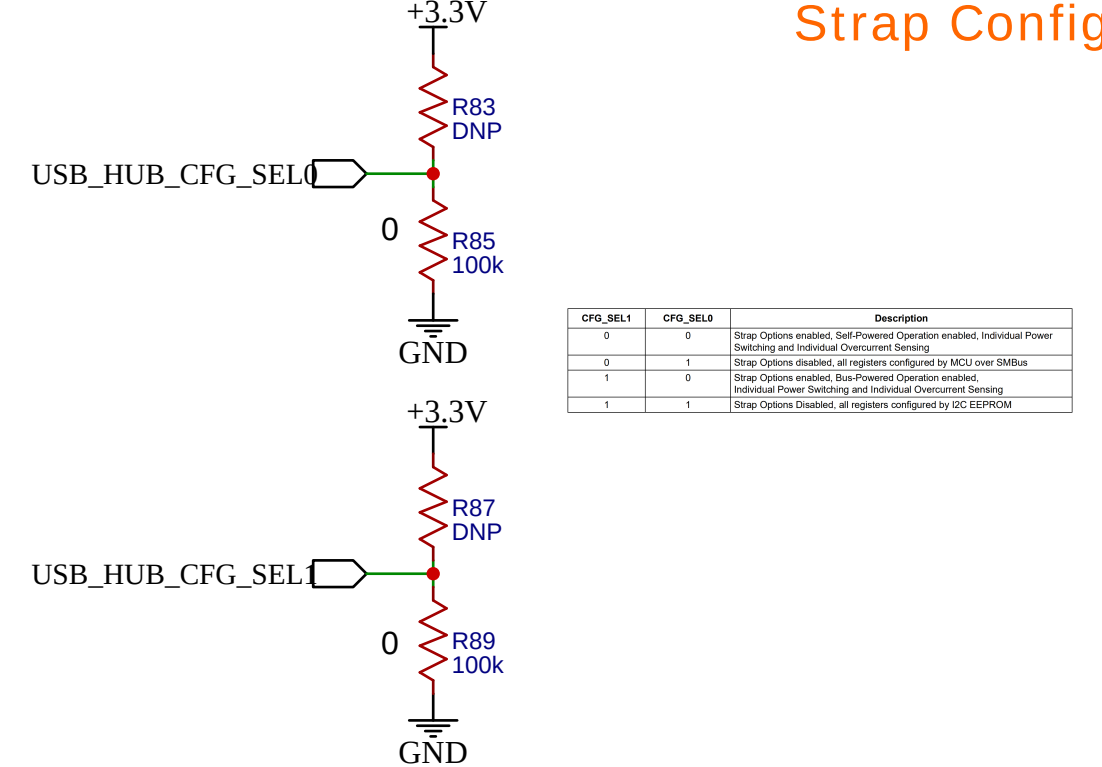
Bus Detection



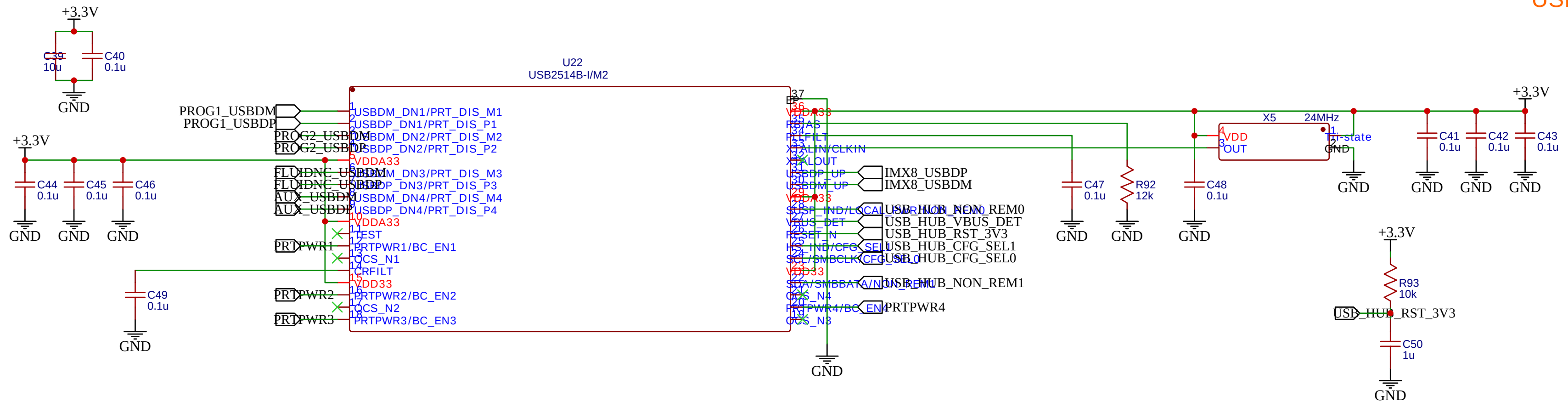
Port Config



Strap Config



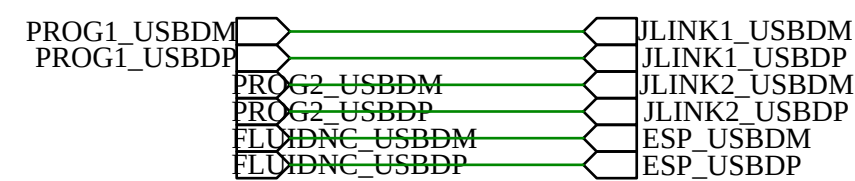
USB Hub



Notes

- USB DM and DP signals already have all necessary USB terminations and resistors in the IC
- VBUS_DET can be attached directly to 3.3V since it is not a detachable hub
- PRTPWR and OCS features are NOT used, as all devices and their power are known
- OCS pins have internal pull ups

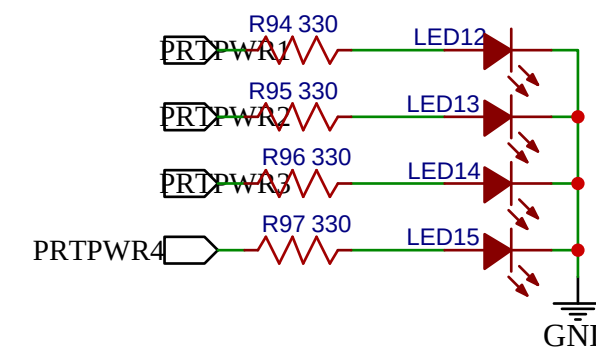
Internal Connections



External Connections



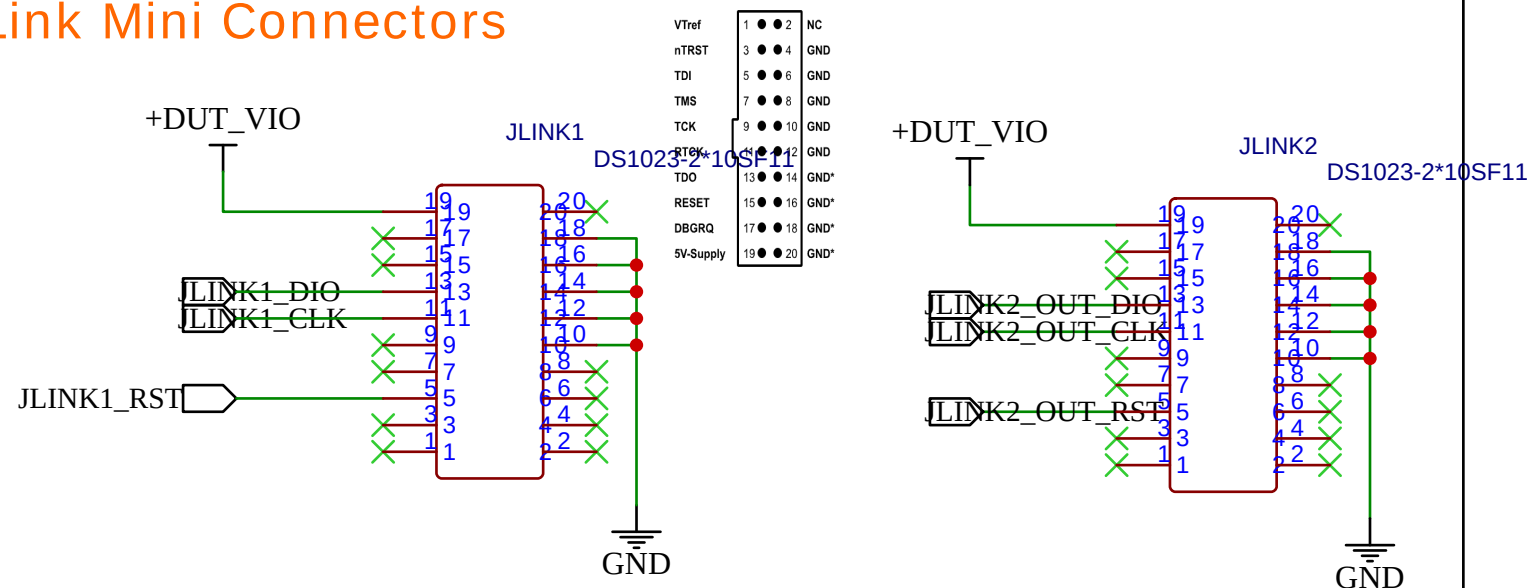
Port Power Debug LEDs



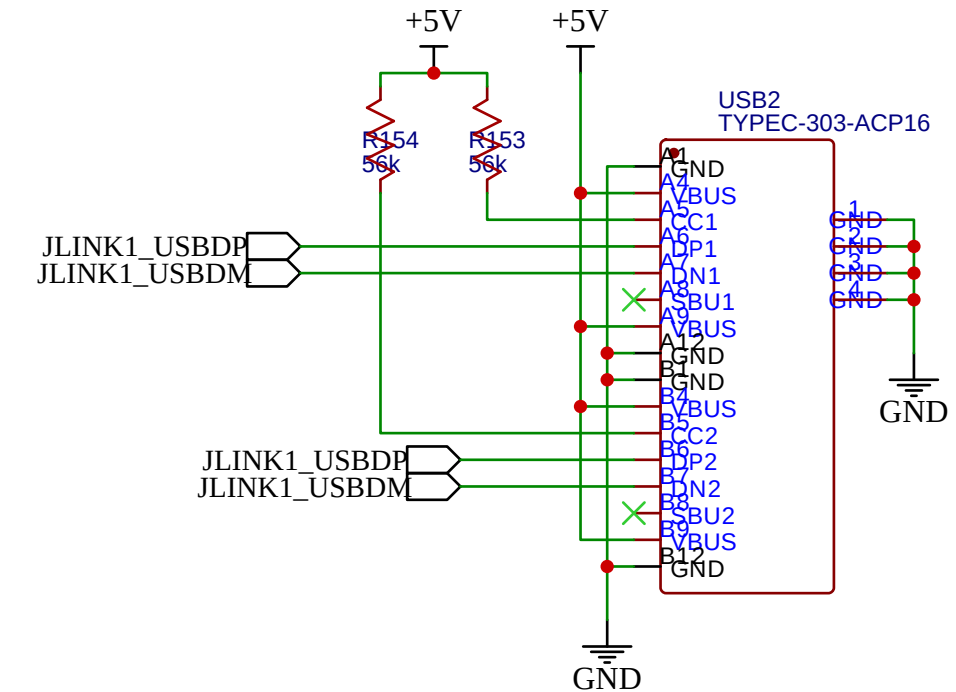
External USB



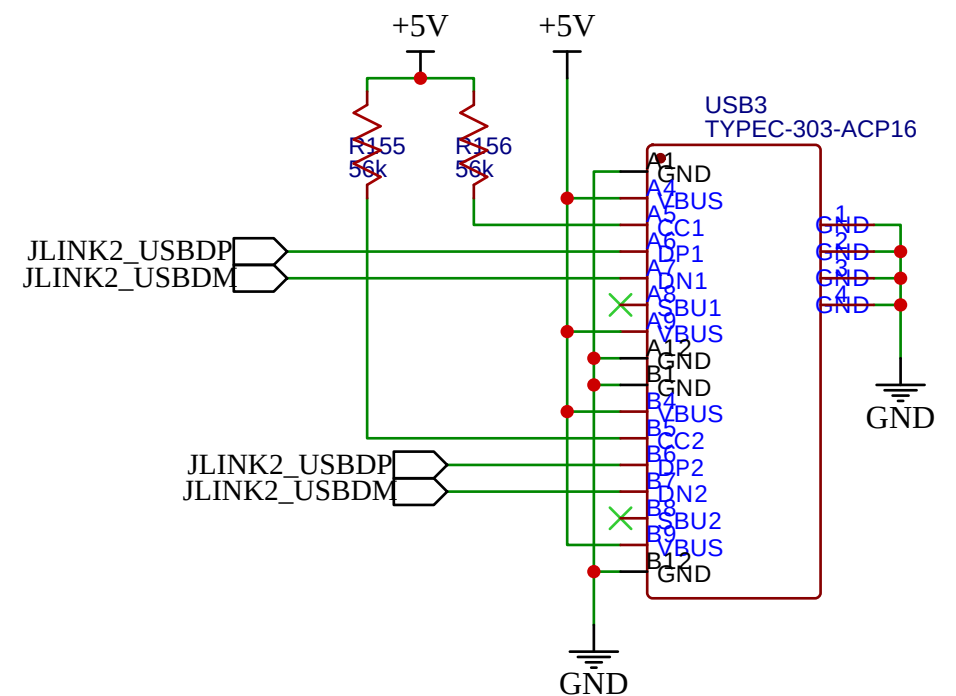
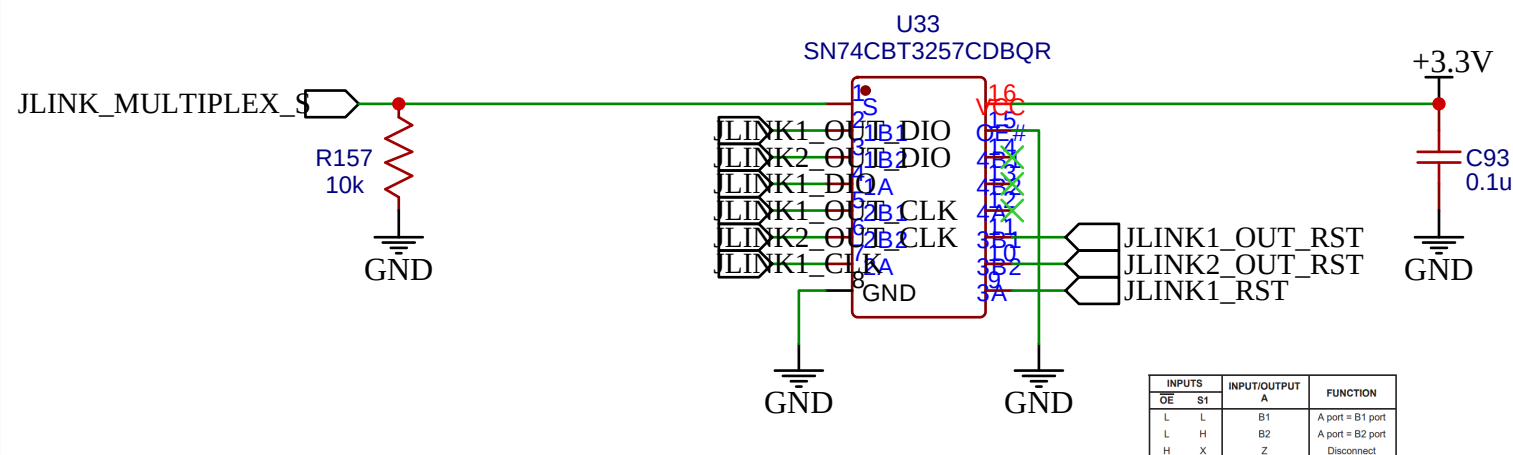
JLink Mini Connectors



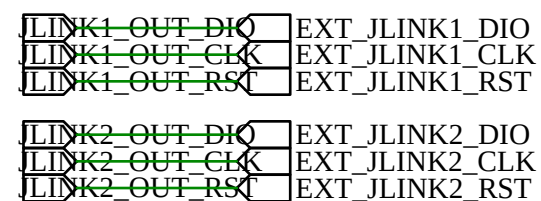
USB-C Source Connectors

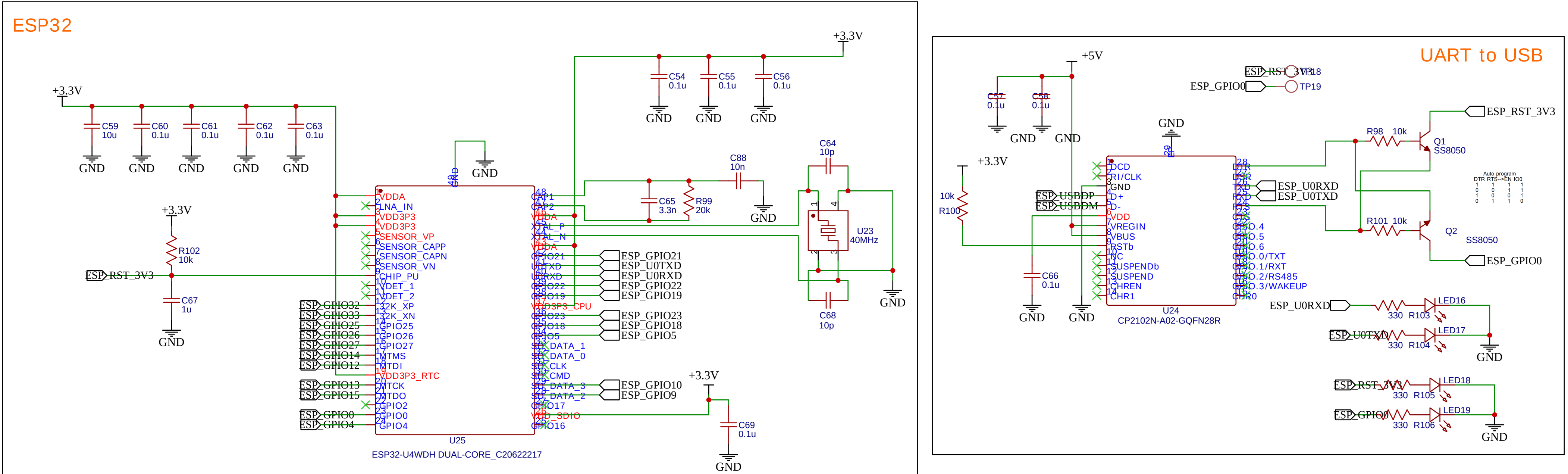


Multiplexer

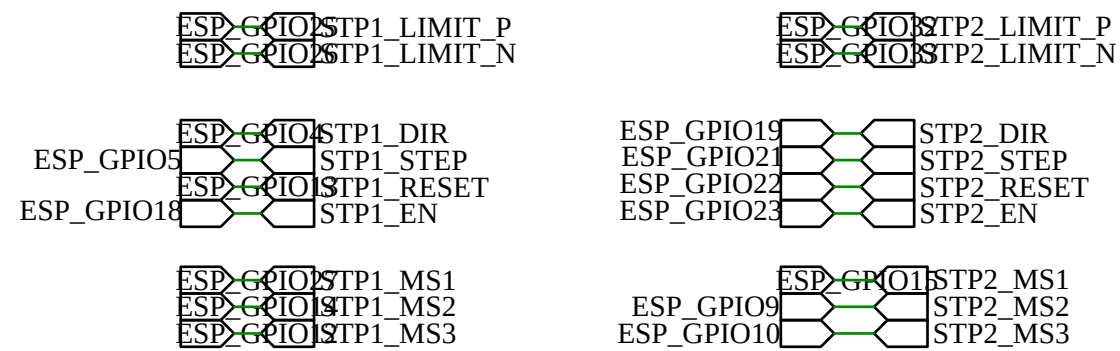


External Connections

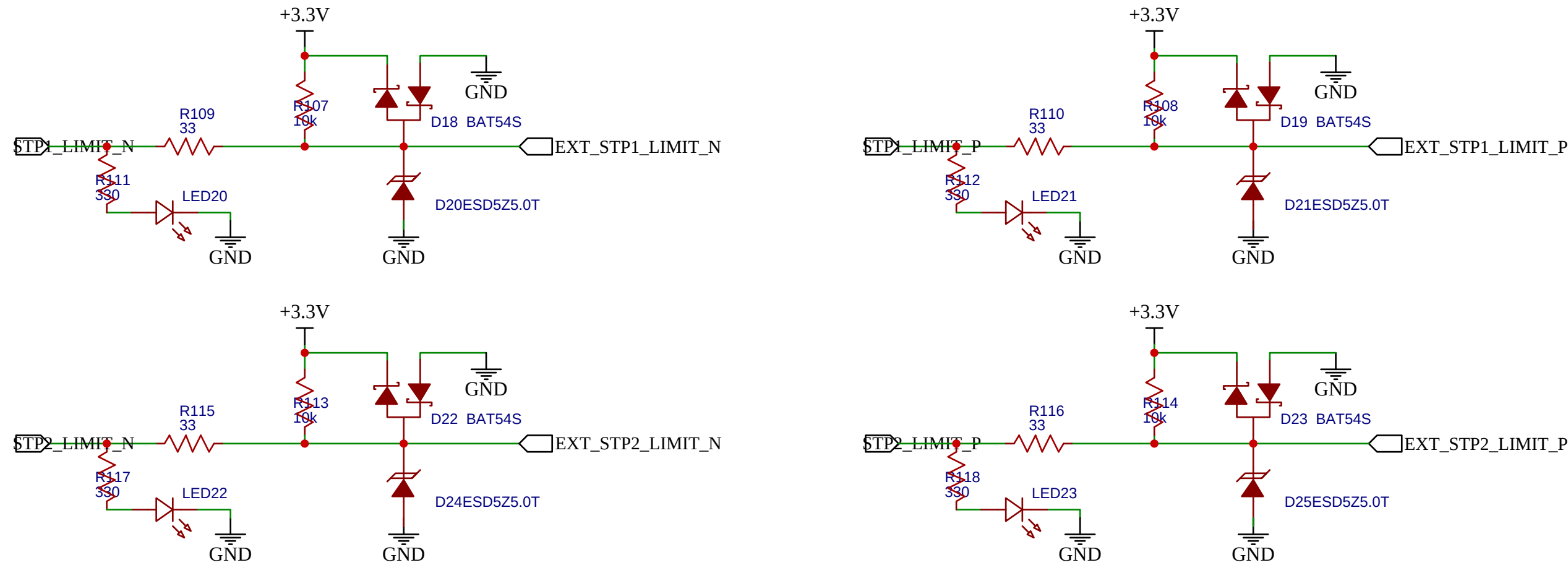




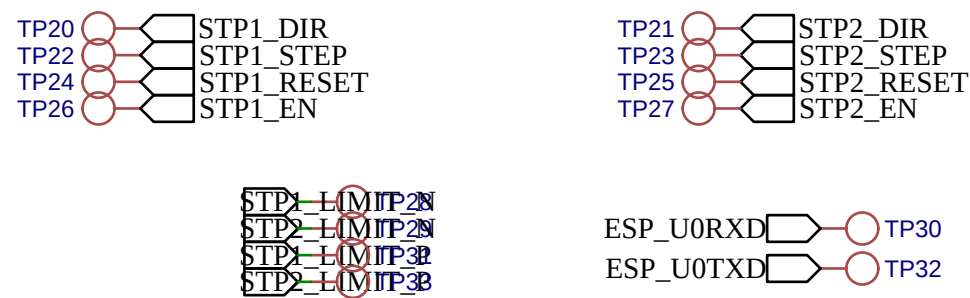
Internal Connections



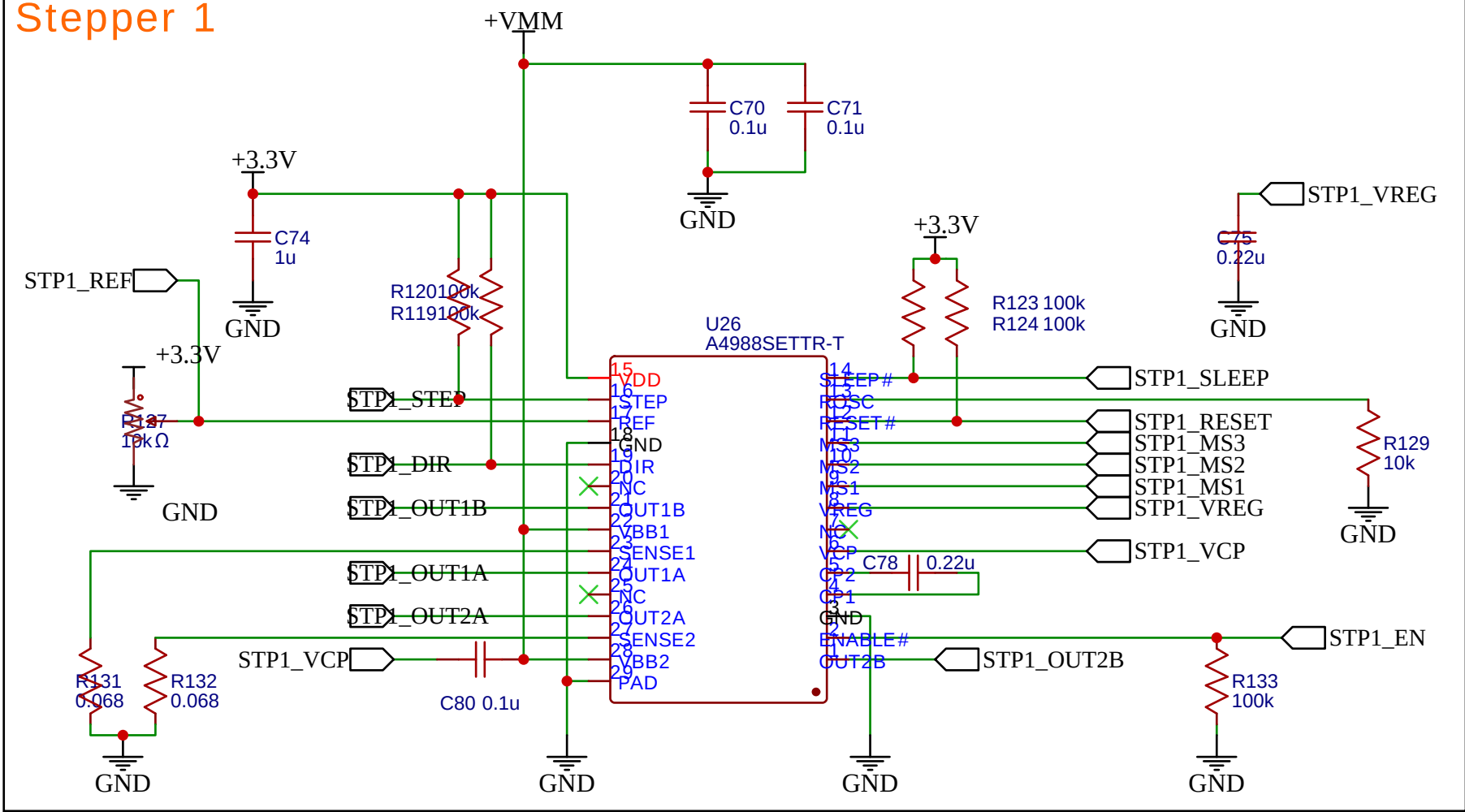
Limit Switches Protection



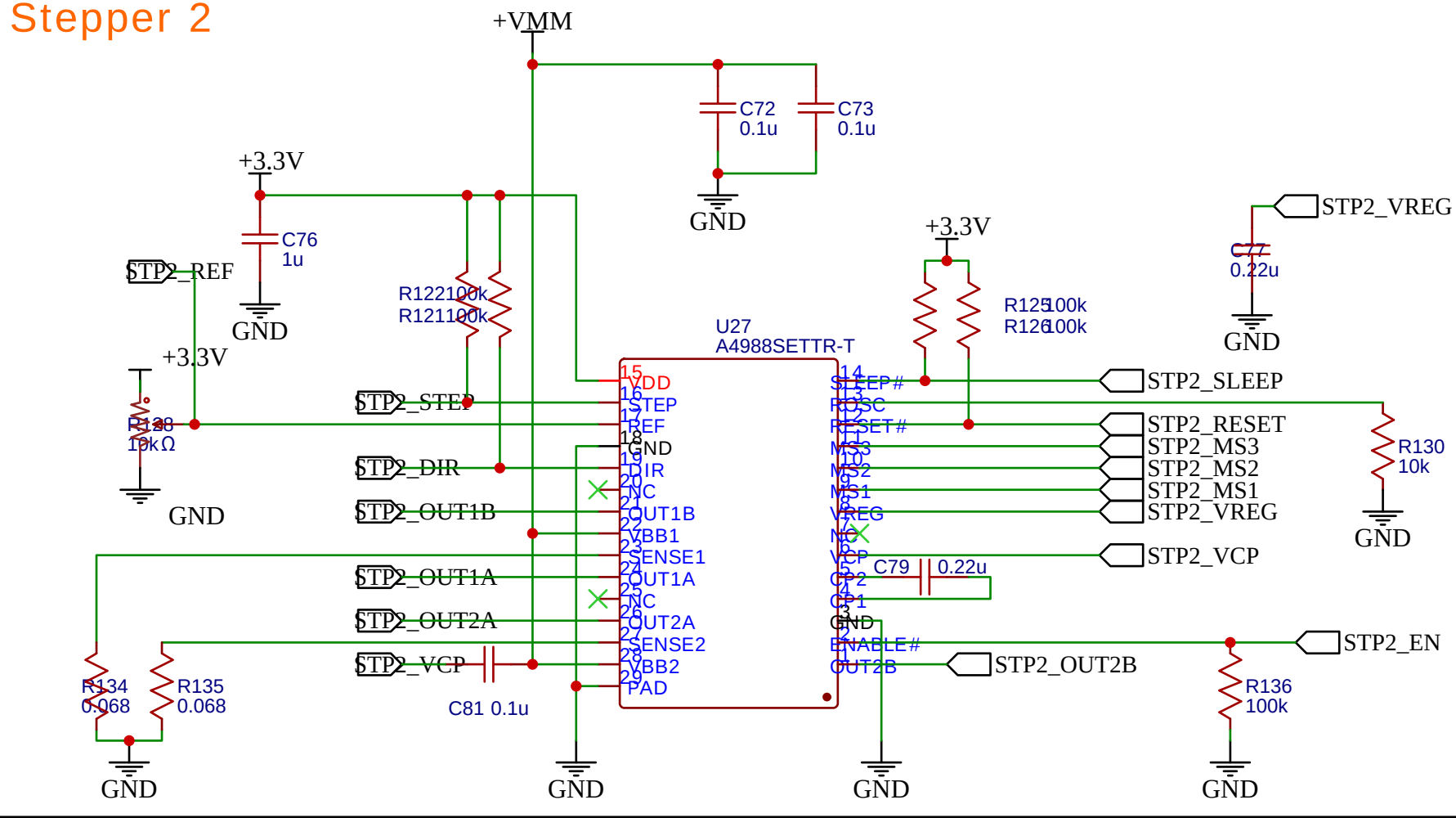
Test Points



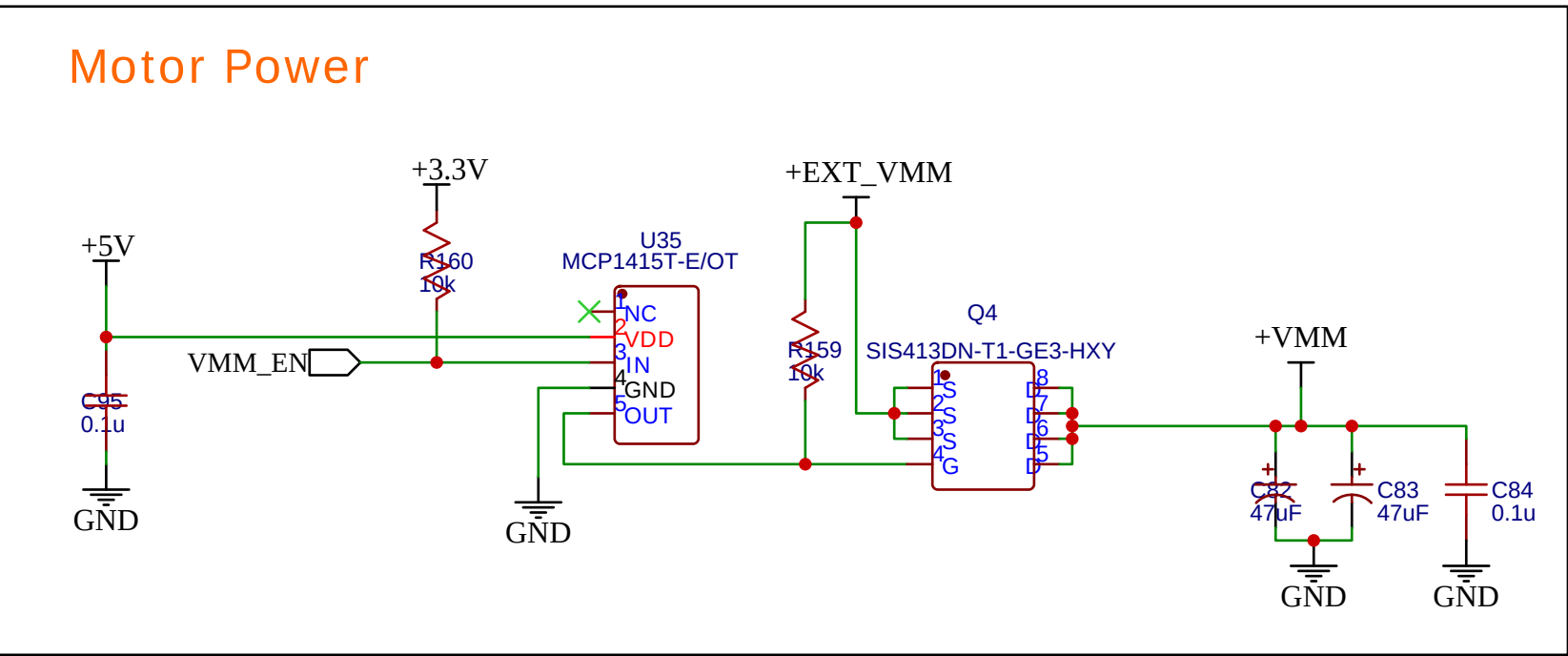
Stepper 1



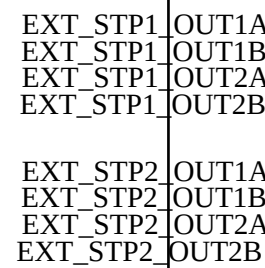
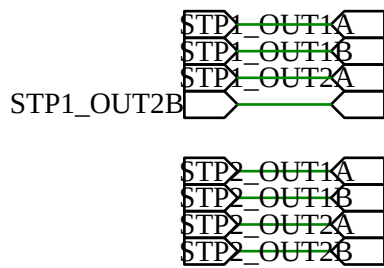
Stepper 2



Motor Power



External Connections

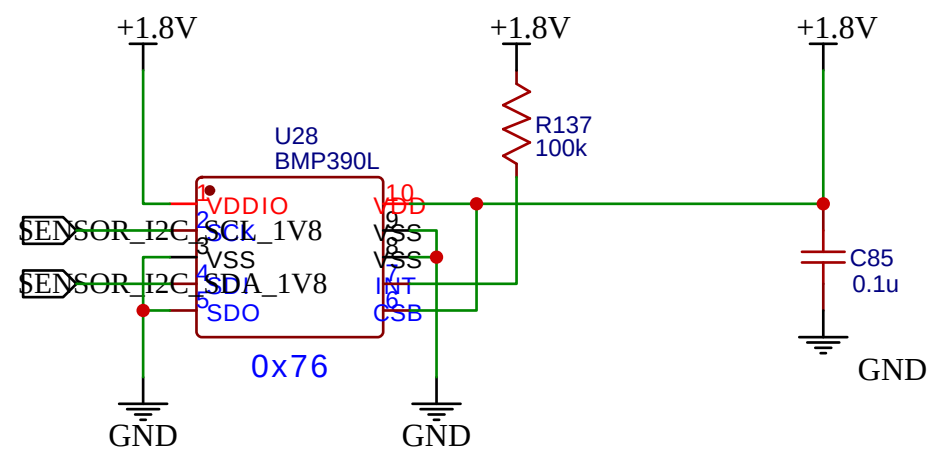


Test Points

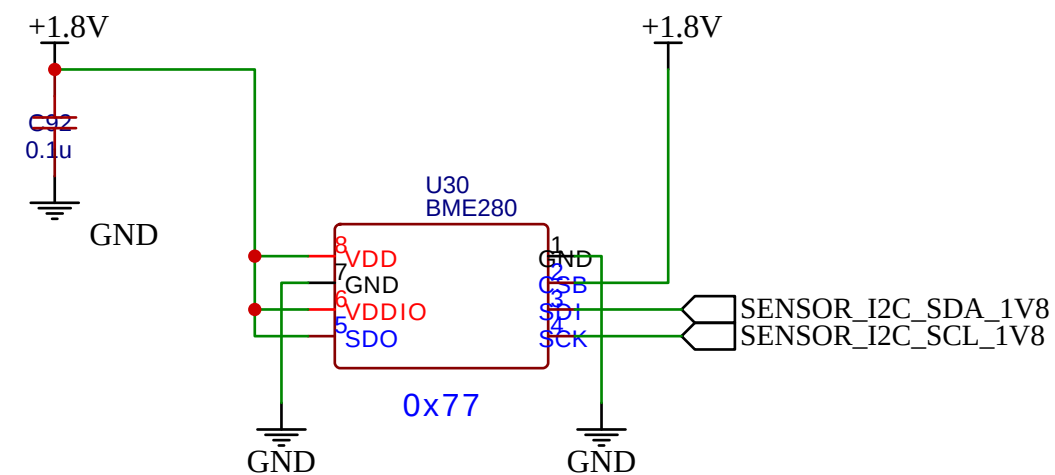
STP2_REF TP40

STP1_REF TP41

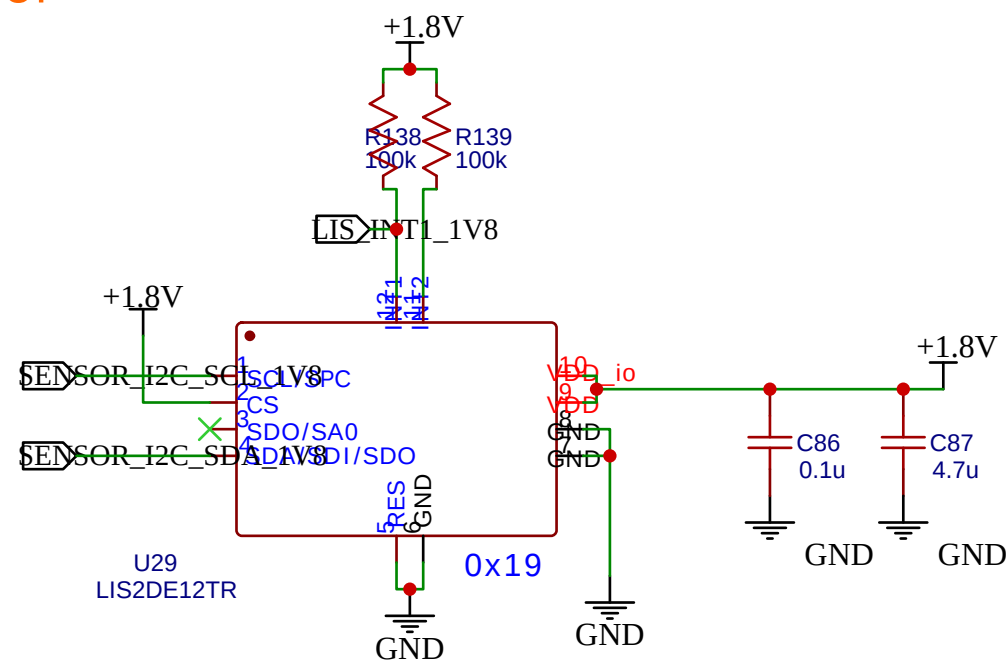
Pressure Sensor A



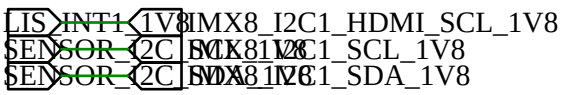
Pressure Sensor B



Accelerometer



Internal Connections



REV1.1

Connectors

- [x] Not all USB cables/connectors have the same polarity
- [x] CN5 should be switched to C588275

Power

- [x] MCP4017T-103E/LT was populated instead of MCP4017T-104E/LT
- [x] TPS Supply needs a 22uF cap on VOUT

ADC

- [x] Replace SPI ADC for 2x I2C ADC
- [x] ADC power LED is too bright
- [x] ADC Signal Conditioning Resistors are not sized correctly
- [x] INA on DUT Power can read voltage, so no need for ADC

Sensors

- [x] Add BME280

FluidNC

- [x] ESP32 pin 48 should NOT be connected to GND. Connect to a 10nF Cap instead

USB

- [x] Set all ports to be "removable" by setting R88 and R84

REV1.2

Connectors

- [x] Fix "qwicc" connector polarity
- [x] Remove unwanted extra UART connectors
- [x] Change recovery and VIO jumpers to sliding switches

Power

- [x] Add motor power switch
- [x] Change digital pot part to more common part number

ADC

- [x] Add 5M Ohm pull downs to ADC channels

Sensors

- [x] Add EEPROM

FluidNC

- [x] Remove fluidnc jumpers

USB

- [x] Change iMX8M USB connection to USB type C (sink)
- [x] Change JLINK ST connectors to USB type C (source)

JLink

- [x] Add JLINK signals multiplexer

Buses

- [x] Flip UART RX/TX lines at level shifter

TITLE: ERRATA		REV: 1.2
COREKINECT	Company: CoreKinect	Sheet:15/15
	Date: 12/05/25 Drawn By: Mateo Segura	